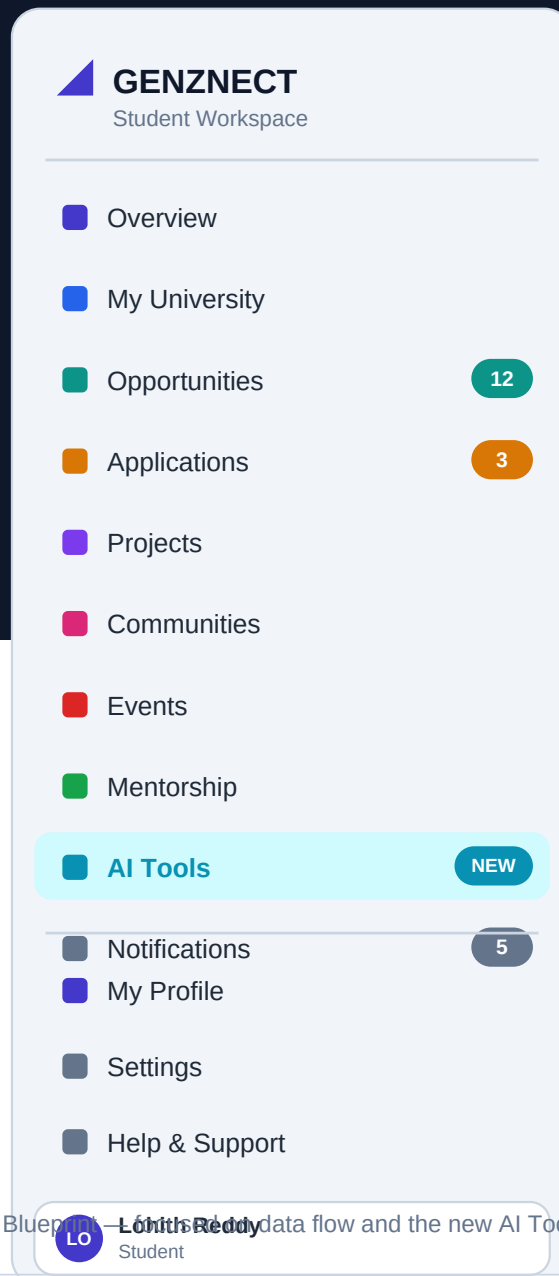


Connections & AI Tools Blueprint

How every module talks to every other module — and reaches the student

Connection map · Update pipelines · Notifications · AI Tools sidebar feature



WHEN "AI TOOLS" IS C

AI Tools

- Resume Build
- Job Search To
- Tools for You

Tools shown adapt
student's registered
e.g. AI/ML → Colab,
Hugging Face · see

Placed just above Noti
personal toolkit rather

Companion to the Student Dashboard Blueprint — Lohith Reddy data flow and the new AI Tools panel

Contents

Every section is listed below and links to its page.

★ Executive Summary	2
1 What “Connected” Means in GenZnect	2
2 Overall Connection Map	2
3 Module Connection Reference	4
4 How Real-Time Updates Reach the Student	4
5 Walkthrough — A New Event Is Published	5
6 Walkthrough — An Application Status Changes	5
7 Walkthrough — A New Opportunity Matches the Student	5
8 Walkthrough — Project Invites & Community Activity	6
9 Notification Channel Matrix	6
10 How Students Connect Their Own Accounts	7
11 New Sidebar Feature: AI Tools	7
12 AI Tools — Where It Lives in the Sidebar	8
13 How Domain-Specific Tools Are Chosen	9
14 Domain → Recommended Tools	9
15 AI Tools Page Layout	10
16 Resume Builder — How It Connects	10
17 Job Search Tool — How It Connects	11
18 Putting It Together — Extended Architecture	11
★ Quick-Reference Connection Table	12

★ Executive Summary

This companion blueprint explains two things in detail: **how every GenZnect module is connected** — what data flows between them and how an update in one place reaches the student — and a **new sidebar feature, AI Tools**, which bundles a resume builder, a job search tool, and a set of tools recommended for the student's own skill domain.

HOW THIS DOCUMENT FITS IN

Nothing here replaces the main Student Dashboard Blueprint. It zooms into the wiring behind it: connections, real-time updates, and the AI Tools panel.

Legend used throughout this document

—————> Automatic data sync - - - - -> Triggers a notification <—————> Two-way connection

1 What “Connected” Means in GenZnect

Every module — My University, Opportunities, Applications, Projects, Communities, Events, Mentorship, Notifications and the new AI Tools — reads from and writes to one shared **Student Data Core**: the student's profile, skills, education, CV and preferences. "Connected" means two things:

- **Data connection** — a module can read the student's profile/skills to personalise what it shows (e.g. Opportunities filters by skill tags).
- **Event connection** — when something changes in one module (a new event, a status change), a notification is triggered and delivered to the places the student actually looks (bell, right panel, module badge).

2 Overall Connection Map

Every module connects back to the same Student Data Core. The arrows run both ways: the core personalises each module, and each module can update the core (e.g. finishing a project adds to the Profile).

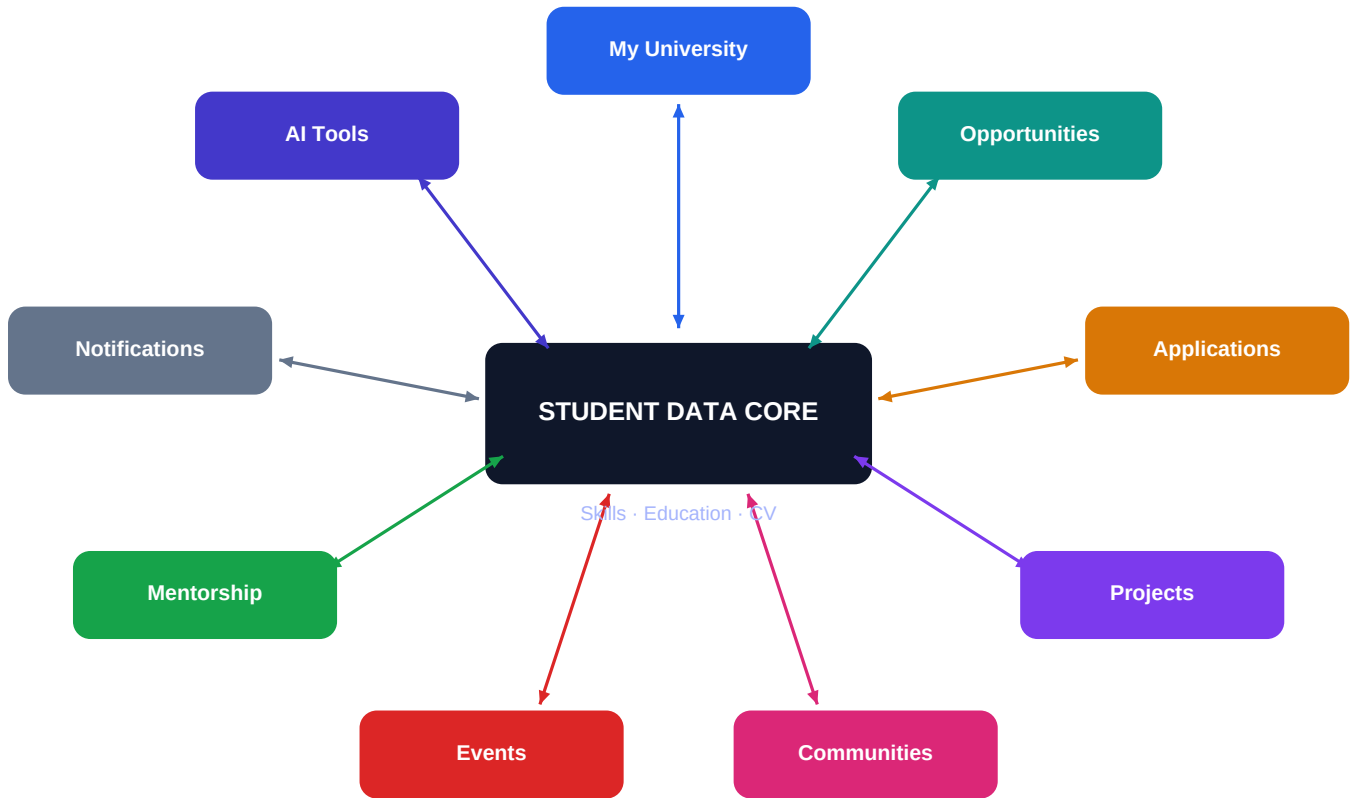


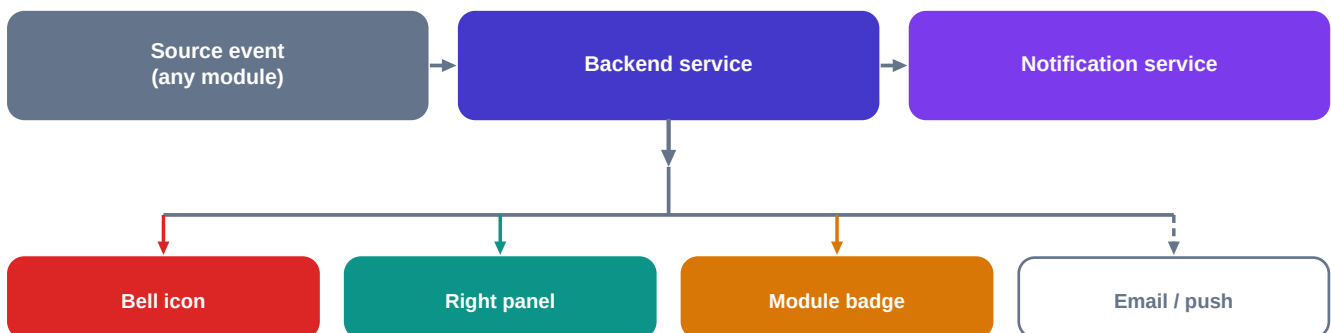
Figure 1 — How every module connects to the Student Data Core

3 Module Connection Reference

Module	Connects to	Data exchanged	Appears in dashboard as
Profile	All modules	Skills, education, CV, preferences, availability	Personalisation everywhere
My University	Profile, Opportunities, Events	Verification status, domain, campus ID	"My University" card, campus-only content
Opportunities	Applications, Profile, Notifications	Postings, required skills, deadlines	Opportunities list, "Recommended for you"
Applications	Opportunities, Notifications	Status, interview schedule	Applications dashboard & timeline
Projects	Profile, Communities, Notifications	Tasks, team, progress	Project workspace, Profile "Projects"
Communities	Profile, Events, Notifications	Posts, memberships, event links	Community feed, "Joined" list
Events	My University, Communities, Notifications	Event details, registration	Events list, "Today" card
Mentorship	Profile, Notifications	Mentor profiles, sessions, ratings	Mentorship page, sessions list
Notifications	Every module	Triggers from any module	Bell, right panel, module badges
AI Tools	Profile, Applications, Opportunities	Skills, resume data, search queries	Resume Builder, Job Search, Tools for You

4 How Real-Time Updates Reach the Student

Whatever the source module, updates follow the same pipeline before they reach the student.



Email / push is optional and follows the student's Settings → Notifications preferences

Figure 2 — The generic update pipeline every module shares

5 Walkthrough — A New Event Is Published

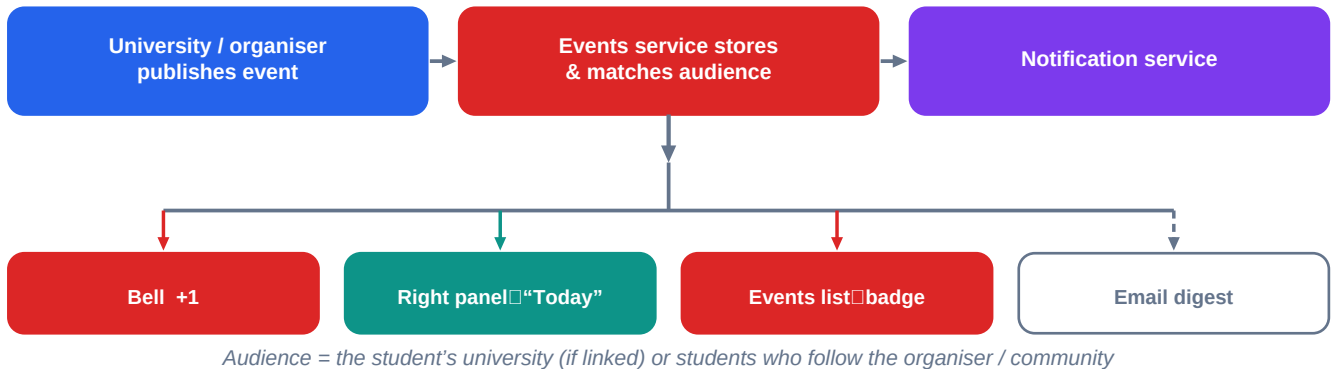


Figure 3 — What happens when a university or organiser publishes an event

This is how the student finds out a new event exists without visiting the Events page: a red +1 on the bell, a card in the right panel's "Today" block, and a fresh badge on the Events sidebar item.

6 Walkthrough — An Application Status Changes

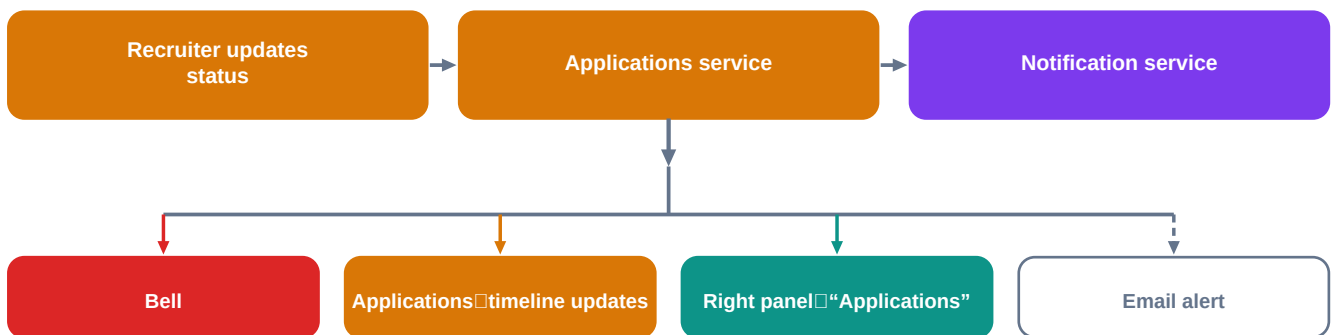


Figure 4 — What happens when a recruiter moves an application forward

The same status change updates the timeline in Section 9 of the main blueprint, the right panel's "Applications" card, and (if enabled) an email alert — all from one update.

7 Walkthrough — A New Opportunity Matches the Student

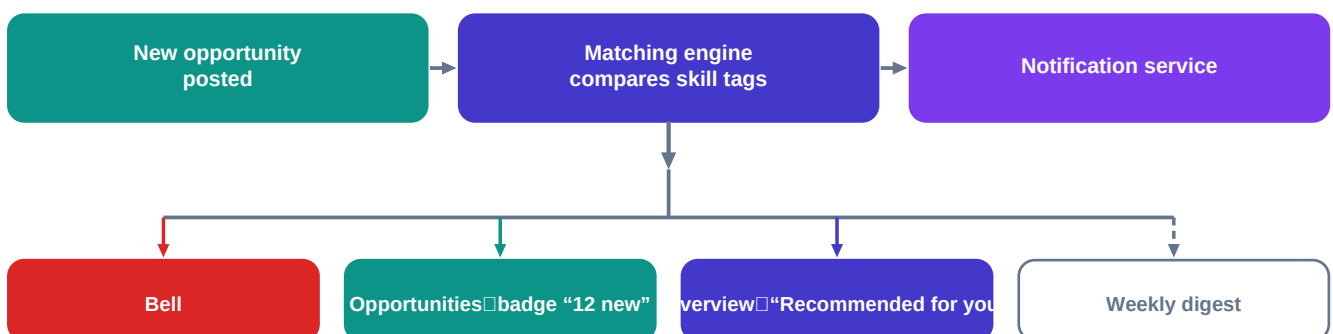
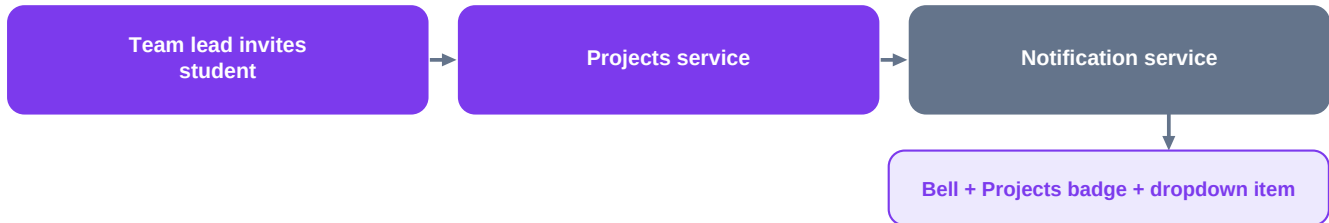


Figure 5 — How a new posting becomes a personalised recommendation

The matching engine compares the posting's required skills against the student's profile skills; a match raises the "12 new" badge on Opportunities and adds a card to Overview's "Recommended for you."

8 Walkthrough — Project Invites & Community Activity

PROJECT INVITATION



COMMUNITY REPLY / MENTION

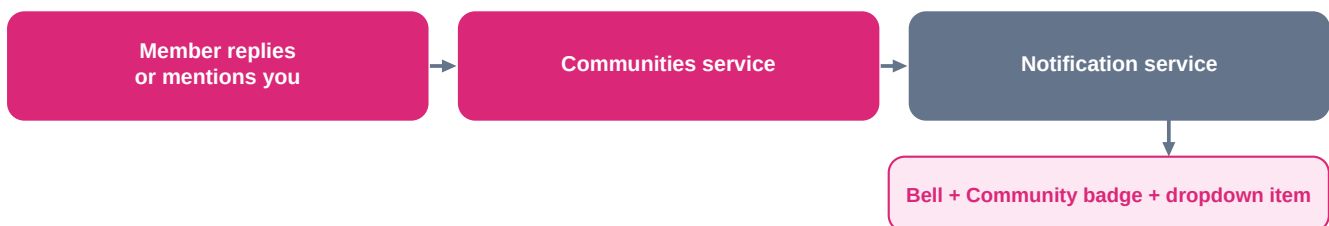


Figure 6 — Two smaller, equally common update flows

9 Notification Channel Matrix

A quick reference for which channels each type of update reaches.

Event type	Bell	Right panel	Module badge	Email / push
New opportunity match	✓	✓ (Overview)	✓	Weekly digest
Application status change	✓	✓	✓	✓ (optional)
New event (campus / global)	✓	✓ (Today)	✓	✓ (optional)
Project invitation	✓	—	✓	—
Community reply / mention	✓	—	✓	—
Mentor session confirmed	✓	✓ (Today)	✓	✓ (optional)
Deadline approaching	✓	✓ (Deadlines)	—	✓ (optional)

10 How Students Connect Their Own Accounts

"Connecting" on the student's side means a handful of simple, explicit actions — each one unlocks or sharpens a part of the dashboard.

What to connect	Where	Steps	Reflects in dashboard as
University account	Settings → University Verification	Enter university email or reference code → verify	"My University" shows Verified; campus content unlocks
Email / push notifications	Settings → Notifications	Toggle channels on or off	Notification delivery matches the chosen channels
Resume / CV	Profile → CV/Resume, or AI Tools → Resume Builder	Upload a file, or build one with Resume Builder	Prefills applications; visible in Profile; feeds Job Search Tool
Skills & career preferences	Profile → Skills / Career Preferences	Add or update skill tags, preferred roles	Powers "Recommended for you," AI Tools domain match, filters
External accounts (future)	Settings → Connected Accounts	Connect LinkedIn / GitHub	Auto-imports skills and projects (planned)

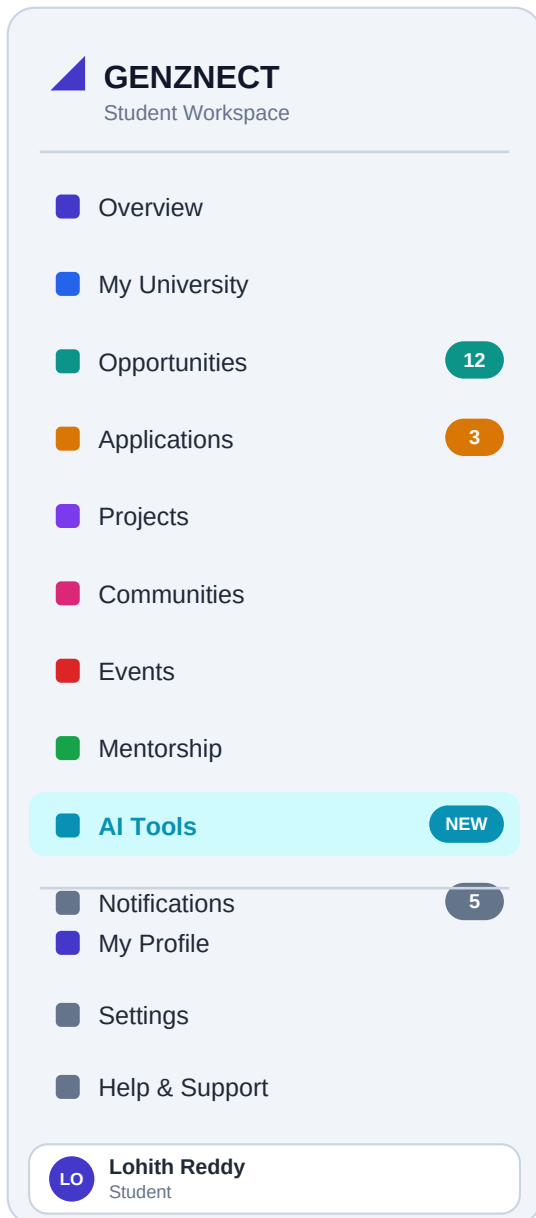
11 New Sidebar Feature: AI Tools

A new sidebar item, **AI Tools**, gives the student a personal toolkit: a resume builder, a job search tool, and a short list of external tools matched to their registered skill set — placed just above Notifications.

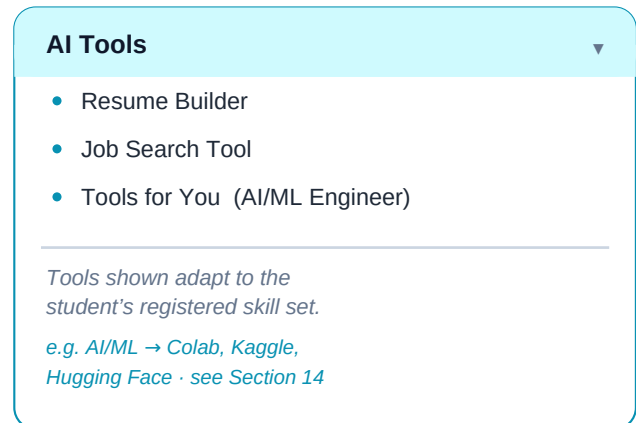
AI Tools reads from the Student Data Core (Section 2) the same way every other module does — it is not a separate app.

12

AI Tools — Where It Lives in the Sidebar



WHEN "AI TOOLS" IS CLICKED



Placed just above Notifications so it reads as a personal toolkit rather than another data feed.

Figure 7 — Sidebar with AI Tools added, and its expanded submenu

13 How Domain-Specific Tools Are Chosen

The tools shown under "Tools for You" are not fixed — they are derived live from the skills in the student's profile.

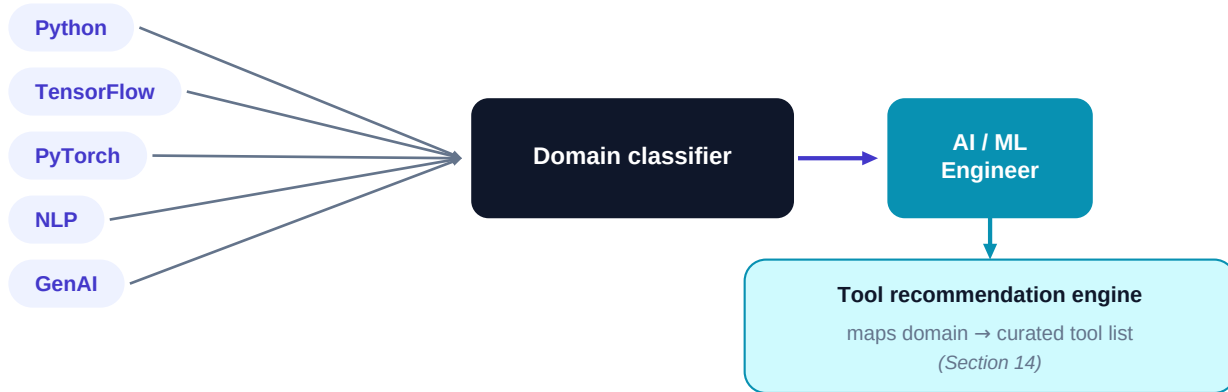


Figure 8 — From skill tags to a recommended tool list

14 Domain → Recommended Tools

Resume Builder and Job Search Tool are always pinned at the top for every student. The grid beneath them changes by detected domain:

Domain (from skill tags)	Example skills detected	Recommended tools shown
AI / ML Engineer	Python, TensorFlow, PyTorch, NLP, GenAI	Google Colab, Kaggle, Hugging Face Hub, Papers with Code, LeetCode (ML), W&B
Java Full-Stack Developer	Java, Spring Boot, React, SQL	Spring Initializr, GitHub, Postman, LeetCode, HackerRank
Data Science	Python, SQL, Pandas, Power BI	Kaggle, Google Colab, Tableau Public, SQL practice sites
Web Development	HTML/CSS/JS, React, Node	GitHub, CodePen, Postman, Vercel
Cybersecurity	Networking, Linux, Pen-testing	TryHackMe, Hack The Box, OWASP resources
General / not yet specified	Any, or profile incomplete	Resume Builder, Job Search Tool, general career resources

Updating skills in Profile refreshes this grid immediately — no separate setup step.

15 AI Tools Page Layout

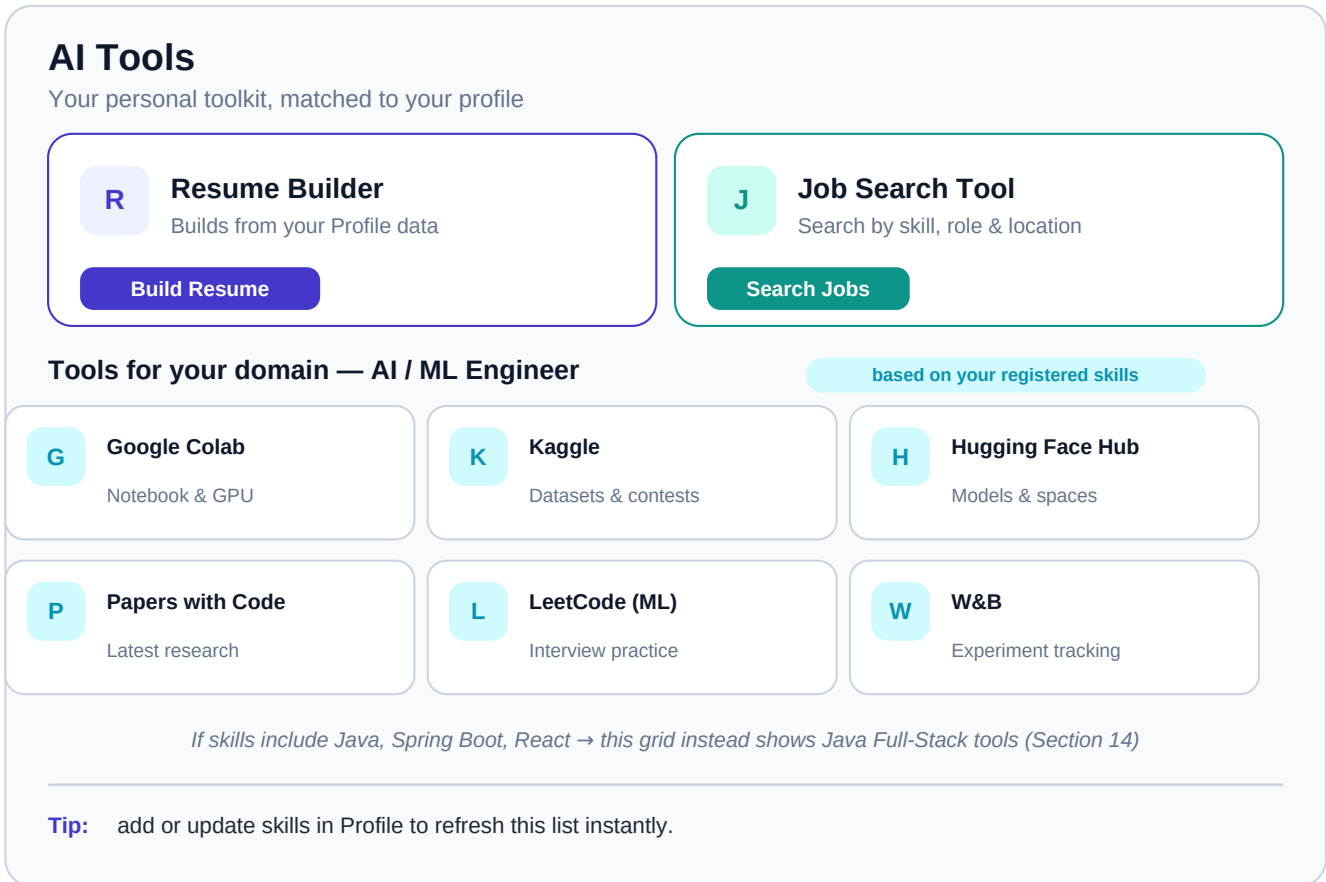


Figure 9 — The AI Tools page, shown here for a student registered as an AI/ML Engineer

16 Resume Builder — How It Connects



Figure 10 — Resume Builder draws from Profile and feeds Applications

The resume is never a separate record to maintain — it is generated from the same Profile data shown in Section 16 of the main blueprint, so updating Profile keeps the resume current.

17 Job Search Tool — How It Connects

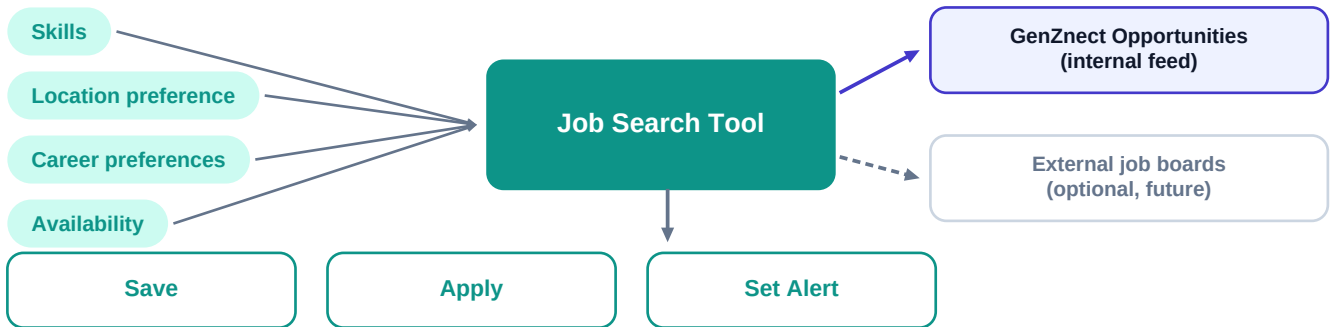


Figure 11 — Job Search Tool blends the internal Opportunities feed with external sources

BUILD ORDER

Ship the internal-feed search first; treat external job boards as a later, optional integration.

18 Putting It Together — Extended Architecture

This extends the main blueprint's architecture with the two additions from this document: AI Tools feeding off Profile, and a Notification Bus that every module reports into.

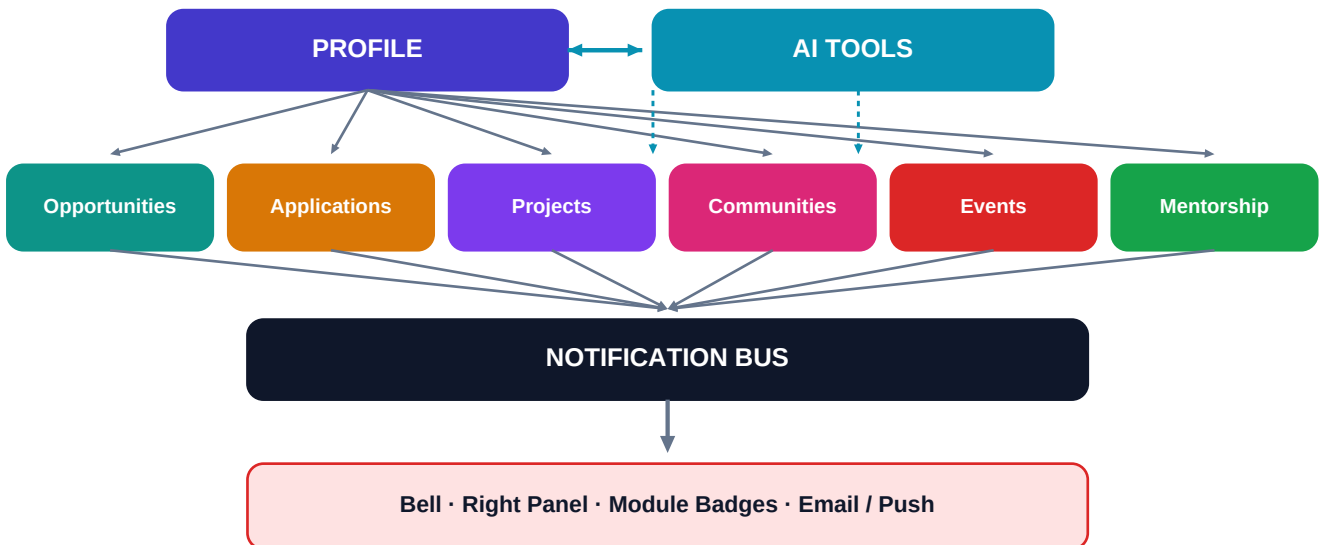


Figure 12 — Extended architecture: AI Tools and the Notification Bus



Quick-Reference Connection Table

One table summarising every connection in this document.

Trigger	Travels through	Lands on
New event published	Events → Notification service	Bell, Right panel, Events badge
Application status changes	Applications → Notification service	Bell, Applications timeline, Right panel
New matching opportunity	Opportunities → Matching engine → Notification service	Bell, Opportunities badge, Overview
Project invitation	Projects → Notification service	Bell, Projects badge
Community reply / mention	Communities → Notification service	Bell, Communities badge
Skill tags updated	Profile → Domain classifier	AI Tools “Tools for You” grid refreshes
Resume requested	Profile → Resume Builder	PDF download, auto-attach to application
Job search run	Profile → Job Search Tool → Opportunities feed	Search results, Save / Apply / Set Alert

DIRECTION

Keep the pipeline in Section 4 as the single source of truth — every new feature should plug into it rather than invent its own notification path.